

Win back up to **R\$ 0.8 M a year** in sales that vanish from the shelf today

The Replenishment Center warns **2.9 days ahead** that an item will run out. The cheapest fix for each case comes ready, and the replenishment manager approves it in **one click**.

TODAY

4.1%

of sales lost to stockouts: R\$ 2.39 M/yr in the sample

CHAIN-WIDE

~R\$ 1.4 M

recoverable per R\$ 100 M of sales (1/3 of the loss)

THE ASK

30 days

pilot in 4 stores, with 4 control stores

The answer on one page

THE PROBLEM

4.1% of sales is lost to stockouts: R\$ 2.39 M a year on just 129 high-turnover items across 20 stores. The team finds out **after** the shelf is empty.

THE VALUE

Up to **⅓ of the loss recovered:** R\$ 0.4 to 0.8 M a year in the sample, depending on execution. Across the chain, **R\$ 0.7 to 1.4 M per R\$ 100 M of sales.**

THE SOLUTION

A model flags **which in-stock items will run out in the next 7 days**. An AI agent checks the facts and proposes the cheapest fix: **transfer, expedite or urgent order**. The manager approves in one click. **All on a single platform, Databricks.**

THE ASK

A 30-day pilot in 4 stores, with 4 control stores. The decision to scale comes from LojaBR's own numbers, not from a projection.

An empty shelf is a sale that goes to the competitor

SALES LOST

4.1%

of sales, to stockouts

PER YEAR · SAMPLE

R\$ 2.39 M

129 items × 20 stores

AVAILABILITY · 7 DAYS

93.5%

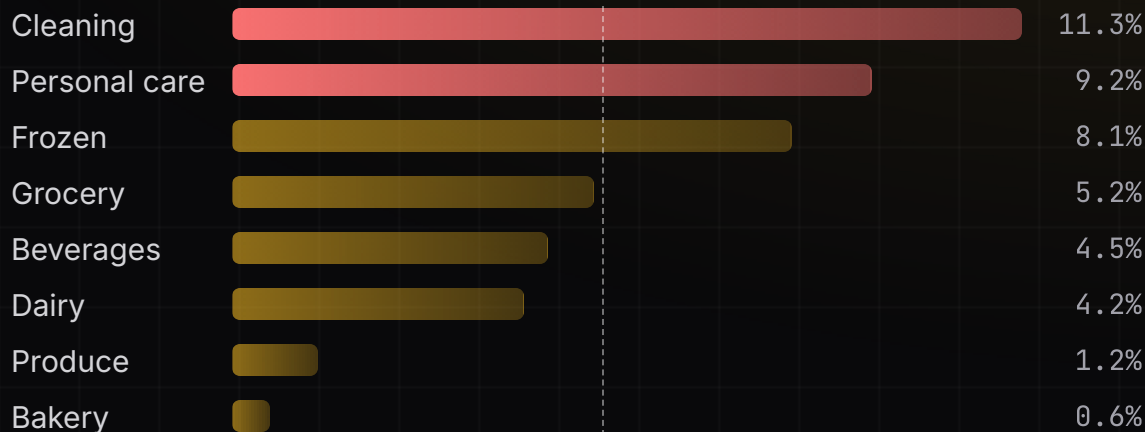
6.5% stockout rate (store × item × day)

WORST SUPPLIER

63%

of deliveries on time

STOCKOUT RATE BY CATEGORY · 120 DAYS · CHAIN AVERAGE 5.3%



The loss is concentrated: cleaning and personal care run out ~2× the average. Their suppliers deliver on time only **63%** and **68%** of the time.

What moves in the numbers LojaBR already tracks

KPI	TODAY	WITH THE REPLENISHMENT CENTER	WHO TRACKS IT
On-shelf availability	93.5%	$\geq 95.5\%$	VP of Operations
Stockout rate (store $\times$ item $\times$ day)	6.5%	$< 4.5\%$	VP and manager
Sales lost to stockouts	4.1% $\cdot$ R\$ 2.39 M/yr	up to $-\frac{1}{3}$ $\cdot$ $\approx$ R\$ 0.8 M/yr back	VP and finance
Warning before the shelf empties	none (reactive)	~ 2.9 days $\cdot$ 75% ≥ 2 days	Replenishment manager
Items to review per day	29% of assortment $\cdot$ 38% right	8% $\cdot$ 84% right	Replenishment manager

Rule of thumb: every **R\$ 100 M of sales** loses $\sim$ R\$ 4 M to stockouts. Recovering a third is worth **$\sim$ R\$ 1.4 M.**

What changes for each of them

EXECUTIVE SPONSOR

VP of Operations & Supply Chain

- **Sales back on the P&L:** $\approx$ R\$ 0.8 M/yr in the sample, $\sim$ R\$ 1.4 M per R\$ 100 M of sales.
- **Idle stock becomes sales:** 39 transfers in a single day used another store's excess, with no new purchase.
- **Risk under control:** the manager decides, and every decision is logged with author and time.
- **Scale without a new project:** the same platform covers the whole chain and new categories.

DOMAIN OWNER

Replenishment Manager

- **A short list:** 8% of items, 3.6 $\times$ fewer than today's rule, sorted by R\$ or by urgency.
- **2.9 days of warning:** time to transfer or expedite, instead of firefighting.
- **The reasoning, ready:** stock, donor store and supplier punctuality on the card itself.
- **One click approves,** and data questions go to Genie in plain Portuguese, no ticket needed.

An action queue, not a report

- 1 Opens the queue at 7 am.** All of the day's actions, sorted by R\$ protected or by what empties first.
- 2 Reads the card.** Product, store, the cheapest fix, the agent's reasoning and the R\$ at stake.
- 3 Approves in one click.** The decision is stored with their name, and **"R\$ protected today"** goes up on screen.

Transfer

from a nearby store
with excess

Expedite

the order already on
its way

Urgent order

when there is no
alternative

CRÍTICO

LIMPEZA

#3

↔ Transferir entre lojas

Sabão em Pó Omo Lavagem Perfeita 1,6kg

Barra da Tijuca (16,3 d) → Tatuapé (0,2 d) 333 km

risco 7 dias
96%

mover
18 un

chegada
2 dias

estoque 1 un · 46 un a caminho · chega em ~4 d · pontualidade do fornecedor 64%

🌟 AGENTE DE REPOSIÇÃO · VERIFICOU 3 FONTES

A loja Tatuapé está com apenas 1 unidade em estoque (cobertura de 0,2 dia) e o pedido a caminho só chega em 2 dias, com fornecedor Limpa Mais tendo apenas 63,6% de pontualidade e atraso médio de 4 dias. A LojaBR Barra da Tijuca tem excedente de 39 unidades e cobertura de 16,3 dias, permitindo transferir 18 unidades sem comprometer seu próprio estoque.

PROTEGE · PRÓXIMOS 7 DIAS

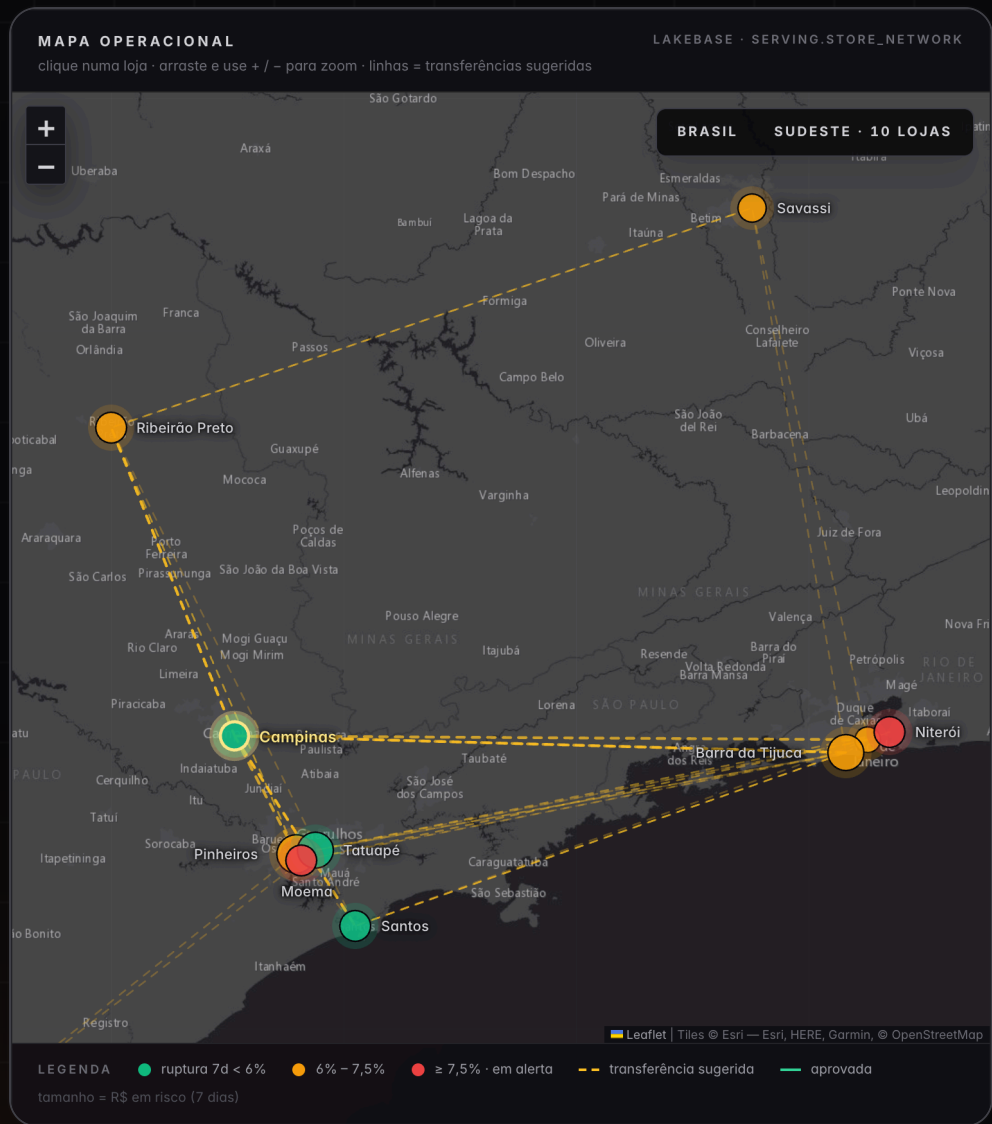
R\$ 516

cobre todo o risco previsto do item

×

✓ APROVAR

WHERE TO ACT FIRST



APP SCREEN · SOUTHEAST ZOOM

Idle stock in one store fixes the gap in another

- **39 transfers** suggested in a single day, using other stores' excess.
- Only within **450 km**, and never leaving the donor with less than **10 days** of stock.
- They arrive in **1 to 2 days**, ahead of the supplier's order.
- The map shows stockouts per store and the routes. One click opens the store and its actions.

"Tatuapé has only 1 unit in stock. LojaBR Barra da Tijuca has a surplus of 39 units, so 18 units can be transferred without compromising its own stock."

excerpt of the agent's reasoning on the previous slide's card (translated)

WHY TRUST THE ALERT

The model is right more than twice as often as today's rule

PRECISION · DAILY LIST

84%

vs 38% for the rule "cover < lead time"

WARNING LEAD

2.9 days

75% of alerts 2 days or more ahead

AUC

0.85

validated on dates the model never saw

LIST SIZE

8%

of in-stock items, vs 29% for the rule

THE AGENT CHECKS BEFORE IT RECOMMENDS

Item position

STOCK · SALES · INBOUND



Nearby stores

EXCESS ≤ 450 KM



Supplier

PUNCTUALITY

If it tries to decide before looking up the facts, the system rejects the decision.

WHAT IT WRITES (TRANSLATED)

"LojaBR Campinas has only 9 units in stock (0.6 days of cover) and the replenishment order only arrives in 2 days. LojaBR Ribeirão Preto, 207 km away, has a surplus of 61 units, so 46 units can be transferred without compromising its own stock."

Café Pilão 500g · evidence run of 2026-09-28

What it is worth, depending on execution

ACTIONS APPROVED AND EXECUTED ON TIME	50 %	75 %	100 %
Sales protected per week · sample	R\$ 8.1 k	R\$ 12.1 k	R\$ 16.2 k
Per year · sample (129 items × 20 stores)	R\$ 0.42 M	R\$ 0.63 M	R\$ 0.84 M
Share of the annual loss recovered	18%	26%	35%
Per R\$ 100 M of chain sales	R\$ 0.7 M	R\$ 1.1 M	R\$ 1.4 M

- **Basis:** one day's queue (run of 2026-09-28) protects R\$ 16.2 k over the next 7 days, $\approx 27\%$ of the previous week's loss.
- **Assumptions:** this is **expected** value (risk-weighted), not guaranteed; the sample is high-turnover items; platform cost (serverless, pay per use) is measured in the pilot.

From raw data to a decision, in one daily job



ARCHITECTURE VIEW IN THE APP (LABELS IN PORTUGUESE) · EVERY STAGE OPENS IN THE WORKSPACE

ONE JOB, EVERY DAY AT 06:00

Ingestion, quality, model, agent and publishing to the app run together, with a single run record.

GOVERNED IN UNITY CATALOG

Lineage for every number, 7 data-quality rules (0 failures in the last run) and least-privilege access for the app.

NO COPIES, NO SPREADSHEETS

The same data feeds the app, Genie and the model. Every stage opens in the workspace straight from the app.

Every number in this deck is traceable: it comes from the daily job's evidence run of 2026-09-28. The executed notebooks, with their outputs, are in the repository as text (evidence/).

What the platform changes in the outcome

	WITH SEPARATE TOOLS	WITH DATABRICKS
Integration	ETL, data warehouse, ML, operational database, LLM, BI and app hosting: 7 pieces to integrate and run	One platform: Lakeflow, Unity Catalog, MLflow, Foundation Model API, Lakebase, Genie and Apps
The number	Copies between systems; the R\$ in the app does not always match the report	One governed dataset: the app, Genie and the model read the same number, with lineage
Time to pilot	Months of integration and one security review per tool	The prototype already runs end to end; the pilot takes 30 days
Security	Permissions and audit spread across every tool	One place: who sees what, and who approved each action
Cost	Fixed licences and infrastructure, running or not	Serverless: the whole journey runs in 8.4 min a day , and only that time is billed

The same data already serves the next use cases:

demand forecasting

markdown of excess stock

store-level assortment

Controlled risk, measured result

WHEN	WHAT HAPPENS	HOW WE MEASURE
Week 1	Connect sales, stock and orders for the 4 pilot stores	data arriving daily, quality $\geq 99\%$
Weeks 2–4	Action queue in the 4 stores; 4 similar stores as control	stockouts and lost sales, pilot vs control
End	Decision to scale	R\$ recovered per store, against cost

SUCCESS CRITERION

Stockout rate in pilot stores $\geq$ **1.5 p.p. below** control, and R\$ recovered greater than the platform cost.

RISKS AND MITIGATION

Model is wrong: the list is ranked by R\$ and the manager decides. Adoption: one click, inside the morning routine. Incomplete supplier data: the agent states its confidence and the rule covers the rest.

Sponsor a **30-day pilot in 4 stores**, with a control group.

Measure stockouts and recovered sales with LojaBR's own numbers, and decide the scale-up on that basis.

WEEK 0 · SPONSOR

Pick the 4 pilot stores and the 4 control stores.

WEEK 1 · LOJABR IT + DATABRICKS

Open the sales, stock and purchase-order feeds.

WEEK 4 · STEERING COMMITTEE

Decide the scale-up with the measured result in hand.

What sits behind each number

- **Lakeflow Declarative Pipelines:** Auto Loader on 7 raw feeds; medallion with 7 data-quality expectations, 0 failures in the last run.
- **Model:** gradient boosting on *point-in-time* features (the open order with its **expected** arrival, not the actual one); AUC 0.85; registered in Unity Catalog via MLflow.
- **Agent:** tool-calling on the Foundation Model API; it queries 3 tools, and the system rejects a decision made before them.
- **Lakebase:** serving schema published atomically and app schema with approvals and the Genie log; reads in ~3 ms.
- **Genie:** entity matching, metric definitions and 5 certified SQL queries.
- **App:** React + TypeScript / FastAPI, service principal with least privilege.
- **Evidence as text:** executed notebooks with outputs, data-quality results, Genie and app responses in evidence/.